

Project

(Due on: May 13, 2026, at midnight)
(Submit on Canvas as one .zip file)

Implement the approach explained in Slide 11 in the attached file “Emotion.pdf”. Data of 3 different subjects is provided. For each subject, you will find in the “Data” folder one .mat files (s0X.mat, where X is the subject number) containing the following attributes:

- **data** (number of trials $\times$ number of channels $\times$ number of samples): Each trial corresponds to a video and contains 10 seconds of EEG signals recorded from 32 channels
- **fs**: Sampling rate
- **labels** (number of trials $\times$ 2):
 - The first column is for valence, and the second column is for arousal
 - 0 for “Low” and 1 for “High”
- **channel_names** (number of channels): EEG channel names.

Deliverables:

- Your code
- For each of the 3 subjects, identify the **frequency band** (δ , θ , α , β , γ), **segment length** (try 0.5 s, 1 s, and 2 s windows) and the value of K (1 to 10) in KNN that yields the highest classification accuracy for both Valence and Arousal. Report the highest accuracy achieved per subject.
- Propose statistical features to be extracted from the time-domain signals and a way to combine features from all channels. Comment on how using all bands affects classification performance per subject.
- Propose a method to combine all frequency bands for each individual channel. Comment on how using all bands affects classification performance per subject.
- Propose a method to combine time-domain features and frequency-domain features. Comment on how using this combination affects classification performance per subject.

CSCE 3611 – Digital Signal Processing

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- Plot the Power Spectral Density for one randomly selected segment of each class (High Valence vs. Low Valence, and High Arousal vs. Low Arousal) using a single EEG channel of your choice. Comment on the visible spectral differences between classes.

Submission:

- Your MATLAB or Python code to be submitted on Canvas on **May 13, 2026, at midnight**.
- A report (to be submitted on Canvas on **May 13, 2026, at mid-night**) that includes the following:
 - Description of the approach used
 - Outputs of the project as described in the deliverables
 - Submission of the above items should be done as one .zip file by the deadline

Guidelines:

- This is a group project. A maximum of 3 students per group is allowed.
- Each team must send an e-mail by **Monday, April 27 at midnight** specifying the members of the team.
- **Changing teams will not be allowed.**
- Project evaluation will occur in the class of **May 14**.
- Project grading will be as follows (out of 15):
 - 5 points on the code submitted
 - 5 points on the submitted report
 - 5 points on the evaluation and discussion